

Introduction and Background

For this project, ArcGIS Pro version 3.5.0 was used to create a suitability model for potential urban agriculture sites in the City of Richmond following the uLESA method proposed by [Hastings et al.](#)¹ uLESA stands for “urban land evaluation and site assessment.” It is an overlay analysis method based on the more rural-focused LESA model developed by the U.S. Department of Agriculture in 1983.² Overlay analysis is a common method for modeling the suitability of a space for a habitat, land use, or site using a diverse set of data sources. uLESA adapts the original model’s method of overlaying maps of land and site characteristics, rating each of these characteristics on a scale from most favorable to least favorable, and producing an overall suitability score.

To account for the specific characteristics that might make a site favorable to agriculture in an *urban* setting, the uLESA method includes the variables food access, investment equity, social vulnerability, and heat island effect. These elements make up the method’s equity and climate resilience submodel. This submodel is combined with more standard agricultural and natural characteristics like slope, environmental impact, contamination, and wetland proximity in its agriculture and natural resources submodel. Both submodels combined provide an overall suitability score that covers the entire Richmond area.

Datasets and Processing Steps

Equity submodel

- Previously collected Virginia Commonwealth data and City of Richmond grocery store data: These data were used to identify [food deserts and food swamps](#) by calculating areas of the city outside of a 10-minute walk buffer from any grocery store or farmers market.³ These areas were given a higher priority score in the equity submodel. Areas within the 10-minute walk buffer received no score.
- American Community Survey (ACS) homeownership and investment survey:⁴ The ACS is an annually-updated survey conducted by the U.S. Census and is publicly available. Results from this Federal survey were used to give higher priority scores to concentrated areas of low income, BIPOC populations, and low rates of BIPOC homeownership as well as areas previously redlined. “Redlining” includes the systematic, historical practice of denying individuals access to credit based on where they lived. Before 1968, when the practice was effectively outlawed, redlining was used to exclude homeowners from securing loans in primarily Black-populated and urban core areas.⁵ The City of Richmond has a documented history of this practice.
- Social vulnerability index: Previously created models developed as a part of the [RVAGreen 2050 Climate Equity Action Plan 2030](#) project were used to identify areas of

the city increasingly vulnerable in the areas of population demographics, health and safety, household make-up, income and education, and housing and transportation.⁶

- Temperature data: These data identified areas of the city that are heat-islands - urban areas experiencing extreme temperatures due to excessive impervious land cover (such as parking lots) and a lack of shade from mature trees. This effect in Richmond was documented by Shandas et al. and published in the journal *Climate* in 2019 as the paper [“Integrating Satellite and Ground Measurements for Predicting Locations of Extreme Urban Heat.”](#)⁷

The scores for food deserts, homeownership and investment, and temperature were weighted by a factor of 2, while the score for social vulnerability was weighted by a factor of 3.

Agriculture and natural resources submodel

- Avoidance of important environmental resources, contaminated sites, cemeteries, and areas with watershed impact: Several datasets were used to exclude any area where urban agriculture might have a significant negative impact, or be otherwise unavailable for use. Boundaries of Resource Protection Areas, contaminated areas, cemeteries, wetlands including a 100 foot buffer, watershed impact areas, and Ecological Cores were all combined for this part of the analysis. Processing of these layers involved an “all or nothing” calculation where any site in one of these areas was automatically excluded from the scoring.
- Soil suitability, slope, and aspect: Three datasets from the USGS and the Virginia DCR were used on a sliding scale of increasing scores based on increasing favorability for a thriving agricultural site. Sites with optimal soil content for growing, less than 10% slope, and orientations allowing for sufficient sunlight exposure were all scored higher than other sites.

All scores in the agriculture and natural resources submodel were weighted by a factor of 1.

For this project, the equity submodel was provided as a pre-processed raster layer. The agriculture and natural resources submodel datasets were geoprocessed in the following steps:

1. New feature added to contaminated sites and cemeteries polygon layer to assign score of -9999 (fail out condition), then this layer was merged with the city boundary layer to indicate areas outside of these sites have a score of 10 (most favorable)
2. Soil suitability polygon later clipped to the city boundary layer, then a new feature was added to contain soil suitability index divided by 10 so that each polygon had a score between 0 and 10
3. Polygon layers delineating Virginia wetlands and ecological cores were clipped to the city boundary layer and a 100 foot buffer added to the wetlands layer
4. These buffered wetland and ecological core layers were merged and dissolved with layers delineating 2030 flood vulnerability and resource protection areas
5. New feature added to this merged environmental resources polygon layer to assign score of -9999 (fail out condition) to all features, then this layer was merged with the city boundary layer to indicate areas outside of these sites have a score of 10 (most favorable)

6. These polygon layers were converted to rasters, a step necessary for further analysis, aligning the coordinate system with and snapping to the pre-processed equity submodel raster layer
7. Watershed impact, soil suitability, land use, and aspect raster layers were reclassified to either assign a score of -9999 to all pixels falling outside of a certain range (fail out condition) or converting a sliding scale of values to the scoring range of 0 through 10; these raster layers were all extracted by mask to the city boundary layer
8. Raster calculator was used to sum the scores of each agricultural and natural resources submodel raster layer
9. Extract by attribute was then used to extract only those pixels receiving a summed score between 0 and 70; these summed scores were converted to a 0%-100% scale where a combined score of 70 = 100% suitability
10. Overall site suitability was calculated by using raster calculator to average the summed scores from the agricultural and natural resources submodel with the summed scores from the equity submodel

Supporting Local Decision-Making

City officials and community members will be able to use this map to both identify potential new sites for urban agricultural projects and to assess proposed sites for suitability. Having all of these important variables together in one composite suitability score precludes the need for users to have in-depth knowledge of urban farming best practices or to perform extensive background research into a city's social equity priority areas.

Map users will also benefit from the stakeholder engagement carried out prior to the overlay analysis development. uLESA developers held stakeholder engagement meetings with city staff, urban agriculture experts, and local community experts to co-create the weighting and analysis methods used for calculating a site's suitability score. This ensures the map is as actionable as possible - stakeholders being engaged from the beginning, rather than after the fact.

Some simple examples of how the map may be used:

- A group of community members come together with an idea for a community garden; prior to engaging their local civic association or city councilmember they use the uLESA map to search their community for the most suitable spaces for urban agriculture
- Staff in the Richmond Office of Sustainability want to encourage more community gardens and urban agriculture in the city; they use the uLESA map to identify the most suitable sites across the entire city and then begin reaching out to local urban farmers to identify interest in expanding to these new sites

Potential for Refinement

This model could be further refined for Richmond by adding information about public or community owned spaces as well as about water access.

Anchoring urban agriculture and community gardens to publicly-accessible sites such as schools, libraries, community centers, and parks could benefit both the public and the garden. The public could take advantage of the environmental benefits, increased food security, and educational value of these gardens. The garden spaces themselves would more likely be maintained and might see greater participation in their upkeep and development if in a higher-traffic, public place within the community.

City-generated and maintained feature layers of [54 RPS schools](#), [9 public libraries](#), and [24 community centers](#) could be added to the equity submodel along with some kind of surrounding buffer (500 feet perhaps).^{8,9,10} A polygon feature layer of [169 City-owned parks](#) could also be added. Pixels falling within these buffer zones or polygons would receive a score of 10 (most favorable).¹¹

Another potential refinement could be adding data about water access to the agriculture and natural resources submodel. While not a requirement, having unimpeded access to a public water source could help urban agriculture projects or community gardens succeed. The city likely has its own maps of water access points that could be used, though these are likely not public. The city does offer public access to a [Service Line Inventory Map](#).¹² The purpose of the map is to identify the material that water lines are made from, however, it also gives an idea of where service lines are located, particularly in park spaces.

Example of water lines identified in a public park:



If it was possible to extract only the service lines within public spaces, these could be buffered to ~100 feet and added as a layer to the agriculture and natural resources submodel. Pixels within 100 feet of a publicly accessible water service line would receive a score of 10 (most favorable).

References Cited

1. Hastings, E., Jones, J. C., Muir, C., Everett, J. C., Ciminelli, J., Llewellyn, J., & Cameron, A. (2025). Adapting the Land Evaluation and Site Assessment (LESA) model for peri-urban and urban agriculture: A geospatial analysis of Chesterfield County, Virginia, USA. *Sustainable Cities and Society*, 131, 106729–106729. <https://doi.org/10.1016/j.scs.2025.106729>
2. Steiner, F., Dunford, R., & Dosdall, N. (1987). The use of the agricultural land evaluation and site assessment system in the United States. *Landscape and Urban Planning*, 14, 183–199. [https://doi.org/10.1016/0169-2046\(87\)90028-4](https://doi.org/10.1016/0169-2046(87)90028-4)
3. *Defining Low-Income, Low-Access Food Areas (Food Deserts)*. (2021). https://www.congress.gov/crs_external_products/IF/PDF/IF11841/IF11841.1.pdf
4. US Census Bureau. (2018, October 11). *American Community Survey Data*. Census.gov. <https://www.census.gov/programs-surveys/acs/data.html>
5. Aaronson, D., Hartley, D., & Mazumder, B. (2021). The Effects of the 1930s HOLC “Redlining” Maps. *American Economic Journal: Economic Policy*, 13(4), 355–392.
6. *Climate Equity Action Plan 2030 Equitable climate action for a healthy and resilient Richmond*. (n.d.). <https://www.rva.gov/sites/default/files/2023-01/FullDocumentRVAgreenClimateEquityActionPlan2030.pdf>
7. Shandas, V., Voelkel, J., Williams, J., & Hoffman, J. (2019). Integrating Satellite and Ground Measurements for Predicting Locations of Extreme Urban Heat. *Climate*, 7(1), 5. <https://doi.org/10.3390/cli7010005>
8. *RPS Schools*. (2025). Arcgis.com. https://richmond-geo-hub-cor.hub.arcgis.com/datasets/b09a218684ad4c39b8681260c9de4959_0/explore?location=37.534819%2C-77.492317%2C11.42
9. *Public Libraries*. (2025). Arcgis.com. https://richmond-geo-hub-cor.hub.arcgis.com/datasets/f9944ea846044fdb11e4c057d2344a7_0/explore?location=37.539904%2C-77.461575%2C11.55
10. *Community Centers*. (2025). Arcgis.com. https://richmond-geo-hub-cor.hub.arcgis.com/datasets/bf13c1b6a172476c87e0e236f6a391b3_0/explore?location=37.529885%2C-77.395168%2C13.67
11. *Parks*. (2025). Arcgis.com. https://richmond-geo-hub-cor.hub.arcgis.com/datasets/3c27cb8526a14b4fa9dd96e928b797a9_0/explore?location=37.533642%2C-77.486177%2C11.25
12. *City of Richmond Service Line Inventory Map*. (2025). Arcgis.com. <https://experience.arcgis.com/experience/ed34aff95e0b4bea8ba11466f1b4796b/>





